

After Lab meeting:

- Confirm research question
- Start group assignment 1
- Download dataset → assign tasks

Start: 2:46 PM

- Simplify histological groups (?) → look at systems of categorization for stages/progression
 - a. Linear progression
 - b. Cluster analysis → to see what are the two groups that stands out (on R its easier to do cause u can visualize it)
 - c.
- We are not going to group by *H. pylori* → +/- *H. pylori*
- Cannot deduce that *H. pylori* has an effect on gastric cancer since there is no control in the paper
- Should still address H.Pylori
 - Take beta diversity (overall microbiome fingerprint)
 - **Additional** analysis
 - Take Gastric cancer + patients → PERMANOVA (does *H.Pylori* correlate with gastric cancer)
 - If significant = control for it
 - If NOT significant = don't control for it
 - Either way we can include in supplemental files to show that we did something you know
 - If it is different -> need to do a control (apparently the formula just changes so not much extra work)
- Question is a little specific in alpha diversity → might want to make the question broader (for group assignment 1) as alpha diversity might be something we want to investigate but other metrics may also want to look into → "Gastric Microbiome"
 - Aim 1: Alpha/beta diversity
 - Aim 2: Taxonomic analysis: differential taxa + functional diversity